Shamit Sai Imandi

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ECE undergraduate at BITS Pilani with a Data Science minor. Passionate about AI/ML systems design, high-performance computing, and graph analytics, with work centered on algorithm optimization, distributed systems architecture, and scalable machine-learning pipelines.

Education

BITS Pilani, Goa Campus **7.31**

B.E. Electronics and Communication Engineering and Minor in Data Science

Experience

Selector AI **May 2026 – July 2026**

AI Engineering Intern

- Built a multi-agent system that automates enterprise network-monitoring onboarding: an orchestrator delegates to a specialist with 46+ live infrastructure APIs to execute a 31-step deployment checklist across inventory, integrations, discovery, alerting, and dashboards.
- Designed the safety layer for agent-driven production changes: 46+ typed APIs cover inventory, device configuration, data pipelines, notifications, and dashboards, while preview/confirm gates require human approval before destructive operations.
- Converted a manual onboarding playbook into configuration-driven agent logic that maps each deployment step to failure modes, remediation tools, dependencies, and required customer inputs; used it to support Major League Baseball (MLB) network-operations onboarding.
- Improved delivery confidence with 165+ automated tests, full-workflow offline simulation, cluster-runtime compatibility fixes, and CI checks for agent routing, tool permissions, and deployment manifests.

SRLDC - Government of Karnataka **May 2025 – July 2025**

Summer Intern Bengaluru, Karnataka

- Built and benchmarked neural-network, gradient-boosted-tree, and ensemble models for next-day grid-demand forecasting; achieved 92% accuracy and outperformed the existing vendor forecast.
- Developed temporal and weather features—including lagged values and rolling statistics—to improve forecast robustness across changing demand patterns.
- Deployed a Linux inference pipeline that reads minute-level PostgreSQL data and produces refreshed rolling forecasts every 15 minutes via cron, enabling continuous 24/7 operation.
- Published forecasts and evaluation metrics to Grafana dashboards, giving operations teams a single real-time view of model performance and demand trends.

Projects

3Daudio *C++17, miniaudio, GLFW, OpenGL, ImGui*

- **Designed a C++17 spatial audio desktop application:** implemented psychoacoustic spatialization techniques, including stereo Vector Base Amplitude Panning (VBAP), Woodworth interaural time delays, and rear-source filtering to simulate distance and direction cues over stereo output.
- **Engineered a callback-driven, 48 kHz float-stereo rendering engine:** processed active synthesizer and decoded audio-file sources per sample in the real-time audio callback; optimized the Digital Signal Processing (DSP) graph using three-band crossovers, fractional delay lines, and parameter smoothing.
- **Simulated room acoustics via custom early reflections:** designed multi-tap floor and ceiling reflection paths utilizing fractional delay lines and specialized low-pass biquad filters to reproduce early-reflection height cues under tight callback deadlines.
- **Developed an interactive 3D graphical editing workstation:** built an isometric visual editor using GLFW, OpenGL, and Dear ImGui for real-time drag-and-drop source positioning and listener yaw controls; synchronized User Interface (UI) state with the audio thread using a shared mutex.

GEMM Kernel *C++20, ARM NEON, cache blocking, multithreading*

- **Accelerated single-precision GEMM throughput by 200×:** improved matrix multiplication performance on 1024^3 matrices from a **1.35 Giga-Floating-Point Operations Per Second (GFLOPS)** naive baseline to **270.55 GFLOPS** in local benchmarks on an 8-core Apple M2 CPU.
- **Outperformed highly tuned hardware-specific libraries:** achieved a **31% performance increase** over Apple M2-optimized, 8-thread Open Basic Linear Algebra Subprograms (OpenBLAS - **206.46 GFLOPS**) by aligning memory blocking parameters with the M2's 12 MB L2 cache hierarchy.
- **Designed a cache-aware memory subsystem and microkernels:** engineered contiguous row-major block packing, 128-byte aligned workspaces with page pre-touching, and hand-unrolled 8×8 ARM NEON Single Instruction, Multiple Data (SIMD) vector microkernels using Fused Multiply-Add (FMA) instructions.
- **Parallelized execution with dynamic task scheduling:** distributed independent output block calculations across CPU cores using 'std::thread' and an atomic tile scheduler; validated numerical correctness against reference loops and passed 2,000-iteration memory stress tests.

Polygraph *Python, SQLite, NetworkX, FastAPI, sentence-transformers*

- **Ingested and managed multi-source streaming datasets:** architected a read-only relational pipeline that harvested **45,520** markets, events, and hierarchical tag structures from public Application Programming Interfaces (APIs), maintaining a structured, local SQLite data store.
- **Engineered an interactive hybrid retrieval system:** combined Full-Text Search 5 (FTS5) with an offline-generated, memory-mapped **384-dimensional vector index (MiniLM - Mini Language Model)**, enabling sub-millisecond keyword and semantic search without query-time embedding re-computation.
- **Built an automated relationship-inference engine:** developed a multi-tier pipeline combining structural constraints, proper-noun entity extraction, semantic similarity, and historical price return correlations (Pearson r) to resolve over **195,650** typed relationships across **38,559** connected markets.
- **Built structured belief landscape visualization:** created an interactive graph explorer that maps how prediction markets interconnect through causation relationships (temporal dependencies, mutual exclusions, correlated movements); users can navigate market influence patterns and understand information flow across 45,000+ nodes using 7-tier relationship inference with confidence scoring.

Achievements

- Joint Entrance Examination (JEE) Mains — 99.1 percentile
- Karnataka Common Entrance Test (KCET) Rank — 734
- Consortium of Medical, Engineering and Dental Colleges of Karnataka (ComedK) Rank — 152
- Joint Entrance Examination (JEE) Advanced Rank — 7811
- 12th Central Board of Secondary Education (CBSE) Boards — 95%
- Codeforces — **Specialist** (1573), lightcf12

Skills

Languages: Python, C/C++, SQL, JavaScript

AI/ML: LLM agents, forecasting, embeddings, graph ML, sentence-transformers

Systems: Linux, ARM NEON, PostgreSQL, SQLite, real-time DSP, multithreading

Tools: FastAPI, NetworkX, Docker, Grafana, Git, CI/CD, OpenGL, ImGui, VBAP